

Supplementary Materials for: “Causal-Role-Aware Extractive Evidence Selection for Power Grid Reliability Reports”

Bijing Liu and Yong Yang

This supplement provides the original candidate-label analyses for provenance (Sections S1–S7) and the new blinded simulated-expert label-sensitivity study (Section S8). Sections S1–S7 must not be interpreted as results on independent human-gold labels; Section S8 contains the revised label views used for the main conclusions.

S1 Supplemental K -Sensitivity Results

Table S1 reports the bounded K -sensitivity check on the original agent-candidate labels. It is retained for reproducibility and is not recomputed on the simulated-expert subset. The primary benchmark budget remains $K = 3$; the additional $K = 1$ and $K = 5$ evaluations were not used to select a new operating point.

Table S1: Historical candidate-label K -sensitivity evidence F1 (best value per column in bold)

Method	$K = 1$	$K = 3$	$K = 5$
TF-IDF query retrieval	0.1783	0.2122	0.2162
BM25 query retrieval	0.1857	0.2273	0.2208
SBERT query retrieval	0.1560	0.1972	0.2042
Full C ² GES reranker	0.2133	0.2983	0.2850

The C²GES advantage is largest and statistically supported at $K = 3$, where the document-cluster bootstrap gain over BM25 is +0.0710 evidence F1 with 95% CI [0.0423, 0.1000]. At $K = 5$, C²GES remains ahead of BM25 by +0.0643 evidence F1 with 95% CI [0.0331, 0.0961]. At $K = 1$, C²GES has the highest aggregate F1, but its paired gains over TF-IDF and BM25 have confidence intervals that cross zero, so the stricter one-sentence setting should be interpreted as suggestive rather than conclusive.

The supplement artifact additionally records paired document-cluster bootstrap comparisons against the semantic baseline at every budget, and these are reported here for completeness. The gain over SBERT query retrieval is +0.0573 evidence F1 (95% CI [0.0093, 0.1013], $p = 0.0182$) at $K = 1$ and +0.0808 ([0.0387, 0.1208], $p < 0.001$) at $K = 5$; at $K = 5$ the gain over TF-IDF query retrieval is +0.0688 ([0.0421, 0.0949], $p < 0.001$). At the primary $K = 3$ budget, the SBERT comparison is significant not only for F1 but also for evidence precision (+0.1067, [0.0683, 0.1467]) and evidence recall (+0.0996, [0.0575, 0.1408]). These additional contrasts come from a single run artifact and are not corrected for multiple comparisons; only the TF-IDF comparison at $K = 3$ retains pre-specified

primary status, and the $K = 1$ SBERT result in particular should be read as diagnostic, since it would not survive a conservative correction over the nine baseline comparisons in the artifact.

S2 Per-Document Dispersion Details

Cross-document heterogeneity under the original candidate labels is substantial. The standard deviations in the archived run artifact are computed over 200 question-level scores and are of the same order as the means (about 0.24 against a mean of 0.2983 for the full reranker), which is expected for a strict top- K set-overlap metric scored per question.

Aggregating to the document level, per-document mean evidence F1 for the full C²GES reranker ranges from 0.0800 to 0.5219 across the 40 documents (mean 0.2983, sample standard deviation 0.1092), with no document at zero. BM25 query retrieval spans 0.0571 to 0.5000, and SBERT query retrieval leaves one document with no recovered evidence at all. Because documents, not questions, are the natural unit of this heterogeneity, all interval estimates in the main text use document-cluster bootstrap resampling (resampling the 40 documents rather than the 200 role questions), since questions from the same document share report context, sentence pools, and annotation history.

S3 Computational Complexity of the C²GES Reranker

The computational cost of C²GES is limited because the method reranks sentences within a single report or excerpt. If a document contains n sentences and text features have effective dimension d , computing query relevance with sparse vectors costs $O(nd)$ in the worst case but is lower under sparse indexing. Role compatibility costs $O(nm)$, where m is the number of role and domain cue features. A naive chain-consistency computation over all sentence pairs costs $O(n^2|\mathcal{R}|)$, but a local context window reduces the cost to $O(nh|\mathcal{R}|)$, where h is the window radius recorded in the implementation configuration. Sorting costs $O(n \log n)$, and the top- K selection can be implemented as $O(n \log K)$ if needed. Since the benchmark contains 40 reports or excerpts and 200 role-conditioned questions, this lightweight design is appropriate for reproducible evaluation without requiring large-scale neural training.

Runtime and hardware are reported for reproducibility, not as a research contribution. Experiments were executed in an environment with CUDA GPU availability on an NVIDIA RTX A6000 with 49140 MB of VRAM. The evaluated methods are lightweight sentence rankers over 40 reports or excerpts, so the computational bottleneck is feature extraction and sentence scoring rather than large model training. This setup makes the benchmark practical to rerun and supports direct inspection of predictions, ablations, and role-conditioned errors.

S4 Hyperparameter Settings and Artifact-to-Table Evidence Map

Table S2 lists the complete experimental settings using short method abbreviations. Abbreviations: Lead = leading-sentence positional selector; TFQ = TF-IDF query retrieval; TFC = TF-IDF centroid salience; TNR = TextRank with NetworkX; LXR = LexRank; CTR = causal trigger ranker; SBQ = SBERT query retrieval; C2G = the full C²GES reranker; QOnly = C²GES query-only ablation; NoRole = C²GES without role compatibility; NoGraph = C²GES without chain consistency.

The SBERT baseline is recorded as the semantic-query condition in the Executor artifact; model checkpoint details are part of the reproducibility package rather than a separate experimental re-

Table S2: Hyperparameter settings

Setting	Value
Task	Role-conditioned evidence-ID selection
Dataset	40 docs, 200 questions
Evidence IDs	608
Roles/doc	5
Budget	$K = 3$
Split	Doc-level 5-fold
Fold rule	SHA256 mod 5
Primary metric	Evidence F1
Main methods	Lead, TFQ, TFC, TNR, LXR, CTR, SBQ, C2G (+ BM25 supplement)
Ablations	QOnly, NoRole, NoGraph, C2G
Selected weights	$w_q = 0.52, w_r = 0.40, w_g = 0.08$
Graph-active roles	Propagation/response, mitigation
BM25 parameters	$k_1 = 1.5, b = 0.75$, lowercase regex tokenization
Label status	125 agent-candidate; 75 AI simulated-expert-adjudicated in revised mixed view
Execution count	1
Hardware	NVIDIA RTX A6000, CUDA, 49140 MB VRAM

sult. TF-IDF baselines use the same sentence and question text normalization as the C²GES lexical query component, and graph-centrality baselines use the same sentence pool as the evidence-ID evaluation.

The released package (<https://github.com/gaoxingkele/c2ges>, release v0.2.0) contains the complete post-freeze supplement artifact (supplement/bm25_k_sensitivity/summary.json) together with machine-derived tables and a cross-check script (code/analyze_supplement_evidence.py). This artifact directly evidences: the main-results rows for TF-IDF query, BM25 query, SBERT query, and full C²GES in Table 3 of the main text (all reported means and standard deviations for F1, precision, recall, and ROUGE-L); the ablation aggregates in Section 5.4 of the main text (query-only 0.2152, no-role 0.2295, no-graph 0.2923) and therefore the ablation mean differences in Table 6 of the main text; the full $K \in \{1, 3, 5\}$ sensitivity results in Table S1; the paired document-cluster bootstrap comparisons of full C²GES against the three query baselines at every K (10000 samples, seed 20260626); the role-stratified values in Tables 4, 5, and 7 of the main text; the per-document dispersion figures in Section S2; and the dataset lineage and verification counts in Section 3 of the main text. The supplement’s independently seeded bootstrap reproduces the main-text bootstrap mean differences exactly; its confidence-interval endpoints for the TF-IDF and SBERT comparisons agree with the printed Executor values to within 0.002, the residual difference reflecting the two runs’ independent bootstrap seeds.

Results not covered by this artifact—the five weak-baseline rows of Table 3 of the main text (lead, TF-IDF centroid, TextRank, LexRank, causal trigger ranker), the confidence intervals and p-values of the four ablation and fixed-legacy comparisons in Table 6, the fixed-legacy condition itself, the cross-validation protocol record, and the qualitative case studies of Section 5.6—are evidenced by the original Executor run artifacts (summary.json, details.jsonl, cv_protocol.json, heldout_predictions.jsonl, metadata.json from the role-selective graph-gate package), which are included in release v0.2.0 under supplement/c2ges_role_selective_graph/. An independent recomputation from details.jsonl (2400 rows, 12 conditions $\times$ 200 questions) verified all 108 manuscript-versus-artifact checks, including the Table 3 means and standard deviations, all Table 6 intervals and p-values, and the cv_protocol.json fold-selection record. The learned and

LLM baseline results of Section 5.5 of the main text are evidenced by the corresponding per-run `summary.json` and `details.jsonl` files under `baseline_runs/` in the same release.

S5 Additional Qualitative Case: Trigger-Event Failure Pattern

A second failure pattern appears in `nerc_012_january_11_oscillation_event_report::trigger_event`. The question asks what event conditions marked the start of the Florida oscillation and activated cyclical turbine valve behavior. The candidate evidence set includes the time-line sentence about the oscillation observed around 08:11:16 UTC and the sentence describing PLI operation causing intercept valves to shut and reopen periodically. Full C²GES selects broader executive-summary and oscillation-description sentences, while TF-IDF retrieves one of the trigger sentences. This shows that the current role cues can prefer general event descriptions over precise trigger-time evidence when the report contains repeated descriptions of the same disturbance.

S6 Paired-Comparison Detail and Run Cost for the Learned and LLM Baselines

This section preserves the learned and LLM baseline comparisons under the original candidate labels. All three baselines were executed with the protocol-matched harness on the same 200 questions, sentence pools, $K = 3$ budget, and metrics; all questions returned valid predictions. Section S8 provides their frozen-prediction rescoring under the revised label views. Both neural rerankers ran locally on CPU; the LLM baseline used the `deepseek-chat` endpoint at temperature 0 with one call per question.

Table S3: Own-score document-cluster bootstrap 95% CIs (evidence F1) and paired comparisons of the learned and LLM baselines (delta = row method – column method; document-cluster bootstrap, 10000 samples)

Method	Own F1 [95% CI]	vs. Full C ² GES	vs. TF-IDF query	vs. SBERT query
BGE reranker (BAAI/bge-reranker-base)	0.2604 [0.2278, 0.2946]	−0.0379 [−0.0739, −0.0010], $p = 0.0448$	+0.0482 [+0.0210, +0.0747], $p < 0.001$	+0.0632 [+0.0260, +0.0995], $p = 0.0006$
Cross-encoder (ms-marco-MiniLM-L-6-v2)	0.2787 [0.2492, 0.3095]	−0.0196 [−0.0539, +0.0140], $p = 0.246$	+0.0665 [+0.0389, +0.0941], $p < 0.001$	+0.0815 [+0.0544, +0.1069], $p < 0.001$
LLM zero-shot (deepseek-chat, temp. 0)	0.5887 [0.5569, 0.6182]	+0.2905 [+0.2533, +0.3253], $p < 0.001$	+0.3765, $p < 0.001$	+0.3915, $p < 0.001$

Both neural rerankers land between BM25 (0.2273) and full C²GES (0.2983), as anticipated for general-domain cross-encoder rerankers applied without domain adaptation, and both beat the lexical and semantic query baselines significantly. Against full C²GES, the BGE deficit is borderline significant at the 0.05 level (the CI upper endpoint is −0.0010), and the cross-encoder deficit is not statistically significant; no significant advantage of C²GES over the cross-encoder is claimed.

Measured LLM run cost and latency: 200 API calls (one per question), roughly 8.5 minutes wall clock (≈ 2.5 s per call including one full-document prompt per call), and an estimated ≈ 0.8 M input tokens across the run (reconstructed from prompt text; the harness client does not record the provider’s usage block), corresponding to approximately US\$0.2–0.3 for the full benchmark at published cache-miss rates—versus zero marginal inference cost for C²GES and local CPU inference for the two neural rerankers. The p -values reported as < 0.001 correspond to bootstrap p -values below the resolution of 10000 samples.

S7 Complete Scoring Specification and Role-Stratified Results

This section preserves the detailed method and original-candidate-label role-wise tables. The scoring definitions were unchanged; revised mixed-label role values are reported in Section S8.

Table S4: Causal roles and representative NERC-style evidence cues

Causal Role	Representative Evidence Cues
Trigger event	fault, trip, outage, line fault, oscillation began, relay operated
Root cause	failed surge arrester, generator step-up transformer, incorrect setting, equipment failure, protection misoperation
Propagation or response	underfrequency, voltage ride-through, relay action, operator response, cascading, automatic control
Impact	load shed, MW reduction, customer interruption, generation loss, equipment damage, net-load increase
Mitigation	restoration, corrective action plan, recommendation, repair, replacement, protection-setting change

Table S5: Complete deterministic scoring specification for the revised C²GES reranker

Component	Definition	Values Used
Query relevance $Q(s, q)$	TF-IDF cosine similarity between each sentence and the role question, followed by min-max normalization	English stopwords; max features 8000; n-grams (1,2)
Role compatibility $R(s, r)$	Cue-hit score plus engineering-unit and domain-abbreviation bonuses, followed by per-role min-max normalization	cue counts 15/21/21/19/19; cue weight $1.0 + 0.15(\ell - 1)$; unit bonus 0.8; abbreviation bonus 0.25
Sentence graph $p(j, k)$	Weighted sequential and semantic proximity between sentence j and neighbor k	$0.45 \exp(- j - k /4.0) + 0.55 \cos_{\text{tfidf}}(s_j, s_k)$; sequential distance capped at 8; semantic edges below 0.08 set to 0
Neighbor roles $A_{r, r'}$	Role-adjacency support used by the chain score	trigger: root/propagation; root: trigger/propagation; propagation: root/trigger/impact; impact: propagation/mitigation; mitigation: impact/propagation
Chain support $C(s_k, r')$	Neighbor’s normalized role score, aggregated through $p(j, k)$	ordered-neighbor weight 1.0; reverse/cross-reference weight 0.45; min-max normalized
Revised graph signal G	Role-gated chain signal	$G = \text{minmax}(R(s, r) \times \text{chain}(s, D, r))$; active only for propagation/response and mitigation
Mixture weights	Document-level 5-fold selected additive weights	$w_q = 0.52, w_r = 0.40, w_g = 0.08$; seven candidates evaluated within folds

S8 Blinded Simulated-Expert Relabeling and Sensitivity Results

A frozen subset of 15 later-added documents was selected before annotation, yielding 75 questions (five causal roles per document). The annotation packet contained the document identifier, role, question, and numbered sentences, but excluded prior answers, prior evidence IDs, notes, method predictions, and performance results. Three separate OpenAI Codex (GPT-5-based) agent

Table S6: Per-role full-C²GES evidence metrics and role-aware deltas

Role	F1	Precision	Recall	vs. TF-IDF	vs. NoRole
Trigger event	0.3114	0.2833	0.4000	+0.1006	+0.0723
Root cause	0.2594	0.2500	0.2896	+0.0470	+0.0475
Propagation/response	0.2611	0.2750	0.2625	+0.0436	+0.0281
Impact	0.3429	0.3500	0.3625	+0.1438	+0.0988
Mitigation	0.3167	0.3167	0.3479	+0.0955	+0.0971

Table S7: Role-stratified evidence F1 at $K = 3$

Role	BM25 Query	NoRole	Full C ² GES	Full – BM25
Trigger event	0.2250	0.2392	0.3114	+0.0864
Root cause	0.2154	0.2119	0.2594	+0.0440
Propagation/response	0.2289	0.2330	0.2611	+0.0321
Impact	0.2181	0.2440	0.3429	+0.1248
Mitigation	0.2490	0.2195	0.3167	+0.0676

instances independently selected one to four minimal direct-evidence sentences or marked a question unanswerable. They followed the same written protocol and did not see one another’s work. These are AI simulated-expert perspectives, not human or domain-expert annotators.

Adjudication used an exact two-of-three sentence majority; a predeclared deterministic fallback was available for questions without a sentence majority but was never needed. The procedure yielded 137 evidence IDs for 75 answerable questions. All-three exact-set agreement was 0.453, any-pair exact-set agreement was 0.867, and 41 questions had at least one set disagreement. Pairwise sentence-level Cohen’s κ was 0.805 (A/B), 0.834 (A/C), and 0.792 (B/C). The original and adjudicated sets matched exactly on 29 of 75 questions (0.387), with mean set F1 of 0.765. Thus the exercise documents substantial sensitivity to label construction.

Predictions were frozen before the blinded annotation and were only rescored; no model was rerun or tuned on the new labels. On the mixed view, full C²GES exceeds no-role by 0.0720 F1 (95% CI [0.0428, 0.1014], $p < 0.001$) and no-graph by 0.0076 ([0.0017, 0.0140], $p = 0.0078$). On the fully replaced subset, its differences versus TF-IDF (+0.0463, $p = 0.0662$), BM25 (+0.0007, $p = 0.9700$), no-role (+0.0078, $p = 0.7610$), and no-graph (+0.0083, $p = 0.2480$) are not significant under document-cluster bootstrap. The subset therefore weakens the method-superiority claim while preserving the benchmark’s value as a provenance and annotation-sensitivity case study.

The released revision directory contains the blind packet and selection manifest; raw A/B/C annotations; adjudicated labels and disagreements; machine-readable agreement summaries; the 15-document fully replaced dataset; the 40-document mixed dataset; and per-question rescored outputs. Original evidence IDs and answers are retained in explicit audit fields rather than overwritten without provenance.

Table S8: Role-stratified evidence-F1 effect of role-selective graph consistency

Role	Full C ² GES	NoGraph	Delta
Trigger event	0.3114	0.3114	0.0000
Root cause	0.2594	0.2594	0.0000
Propagation/response	0.2611	0.2527	+0.0083
Impact	0.3429	0.3429	0.0000
Mitigation	0.3167	0.2952	+0.0214

Table S9: Frozen-prediction results under two revised label views

Method	Mixed 200		Replaced subset 75	
	F1	Hit@3	F1	Hit@3
TF-IDF query retrieval	0.1987	0.4750	0.2257	0.4933
BM25 query retrieval	0.2213	0.5150	0.2712	0.5733
SBERT query retrieval	0.1851	0.4500	0.2321	0.5067
C ² GES query-only	0.2012	0.4850	0.2488	0.5467
C ² GES without role	0.2161	0.5200	0.2641	0.5733
C ² GES without graph	0.2805	0.6350	0.2637	0.5600
Full C ² GES	0.2882	0.6600	0.2719	0.5867
BGE reranker	0.2500	0.5650	0.3035	0.6667
Cross-encoder	0.2717	0.6100	0.3155	0.6667
LLM zero-shot	0.5872	0.9550	0.5699	0.9333

Table S10: Mixed-label role-stratified F1 and Hit@3 for full C²GES

Role	F1	Hit@3	F1 vs. BM25	F1 vs. NoRole
Trigger event	0.3181	0.650	+0.0711	+0.0694
Root cause	0.2394	0.575	+0.0582	+0.0700
Propagation/response	0.2308	0.625	+0.0164	+0.0309
Impact	0.3311	0.725	+0.1140	+0.0855
Mitigation	0.3214	0.725	+0.0747	+0.1043